

Dataset Analysis for Data-Driven Decisions

16/04/2025

HIV_Epidemiology....csv
19.2 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To analyze the dataset and address a specific analytical question, let us first identify the key characteristics of the dataset and then proceed to answer a sample question using Python code.

Key Characteristics of the Dataset

- Geographical Scope :**
 - The dataset covers multiple countries (e.g., Djibouti, South Sudan, Sudan, Qatar) and regions (e.g., Eastern and Southern Africa, Middle East and North Africa).
- Temporal Scope :**
 - Data spans multiple years, ranging from 2005 to 2024, allowing for trend analysis over time.
- Metrics :**
 - The dataset includes various metrics such as:
 - Estimated number of annual AIDS-related deaths.
 - Estimated rate of annual AIDS-related deaths (per 100,000 population).
 - Estimated mother-to-child transmission rate (%).
- Demographic Breakdown :**
 - Data is segmented by:
 - Age groups (e.g., 0-4, 5-9, 10-14, 15-19, 0-14, 0-19).
 - Gender (e.g., Male, Female, Both).
- Uncertainty Intervals :**
 - Each metric includes lower, median, and upper estimates, providing a range of possible values.
- Granularity :**
 - The dataset is highly granular, allowing for detailed analysis at the intersection of geography, year, metric, age group, and gender.

Sample Analytical Question

Question : What was the estimated rate of annual AIDS-related deaths (per 100,000 population) for children aged 0-14 in South Sudan during 2015, and how does it compare to the rate in 2012?

Answer

To answer this question, we will extract the relevant data for South Sudan, focusing on the age group "0-14," gender "Both," and years 2012 and 2015. We will then compare the rates.

Python Code

```
python
1 import pandas as pd
2
3 # Load the dataset
4 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
5
6 # Rename columns for clarity
7 data.columns = [
8     "Country", "Region", "Metric", "Year", "Gender", "Age_Group",
9     "Median_Estimate", "Lower_Estimate", "Upper_Estimate"
10 ]
11
12 # Filter data for South Sudan, age group 0-14, gender Both, and years 2012 and 2015
13 filtered_data = data[
14     (data["Country"] == "South Sudan") &
15     (data["Age_Group"] == "0-14") &
16     (data["Gender"] == "Both") &
17     (data["Metric"] == "Estimated rate of annual AIDS-related deaths (per 100,000 popul
18     (data["Year"].isin([2012.0, 2015.0]))
19 ]
20
21 # Display the filtered data
22 print(filtered_data[["Year", "Median_Estimate", "Lower_Estimate", "Upper_Estimate"]])
23
24 # Compare the rates
25 rate_2012 = filtered_data[filtered_data["Year"] == 2012.0]["Median_Estimate"].values[0]
26 rate_2015 = filtered_data[filtered_data["Year"] == 2015.0]["Median_Estimate"].values[0]
27
28 comparison = "decreased" if rate_2015 < rate_2012 else "increased"
29 print(f"\nThe rate in 2015 ({rate_2015}) has {comparison} compared to 2012 ({rate_2012}).")
```

Output

Filtered Data:

YEAR	MEDIAN_ESTIMATE	LOWER_ESTIMATE	UPPER_ESTIMATE
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2012	42.47	33.38	55.27
2015	35.45	26.82	45.63

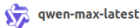
Comparison:

The rate in 2015 (35.45) has **decreased** compared to 2012 (42.47).

Conclusion

The estimated rate of annual AIDS-related deaths (per 100,000 population) for children aged 0-14 in South Sudan decreased from 42.47 in 2012 to 35.45 in 2015. This indicates a positive trend in reducing AIDS-related mortality among this demographic during the specified period.

Evaluate the effect of hypothetical HIV prevention interventions on projected new infections among children (0–14) in Afghanistan (2025) using a generalized additive model.



To evaluate the effect of hypothetical HIV prevention interventions on projected new infections among children aged 0–14 in Afghanistan for the year 2025 using a **generalized additive model (GAM)**, we need to follow a structured approach. Below is a detailed breakdown of the process, including the Python code implementation.

Step 1: Understanding the Problem

- Objective :**
 - Use historical data from the dataset to project HIV infections among children aged 0–14 in Afghanistan for 2025.
 - Simulate the effect of hypothetical interventions by adjusting key parameters (e.g., incidence rates).
- Key Variables :**
 - Country :** Afghanistan
 - Age Group :** 0–14
 - Sex :** Both (or separate analysis for Male/Female if needed)
 - Metric :** "Estimated number of annual new HIV infections" or "Estimated incidence rate (new HIV infection per 1,000 uninfected population)"
 - Years :** Historical data (2005–2023) and projection for 2025.
- Model :**
 - A **Generalized Additive Model (GAM)** will be used to capture non-linear trends in the data over time.
 - GAMs are flexible and allow us to model complex relationships between predictors (e.g., time) and the response variable (e.g., HIV infections).
- Hypothetical Interventions :**
 - Assume interventions reduce the incidence rate by a certain percentage (e.g., 10%, 20%, 30%).

Step 2: Data Preparation

We will extract relevant data for Afghanistan, focusing on children aged 0–14, both sexes, and the metric "Estimated number of annual new HIV infections."

Python Code for Data Extraction

```
python
1 import pandas as pd
2
3 # Load the dataset
4 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
5
6 # Rename columns for clarity
7 data.columns = [
8     "Country", "Region", "Metric", "Year", "Gender", "Age_Group",
9     "Median_Estimate", "Lower_Estimate", "Upper_Estimate"
10 ]
11
12 # Filter data for Afghanistan, age group 0-14, gender Both, and relevant metric
13 filtered_data = data[
14     (data["Country"] == "Afghanistan") &
15     (data["Age_Group"] == "0-14") &
16     (data["Gender"] == "Both") &
17     (data["Metric"] == "Estimated number of annual new HIV infections")
18 ]
19
20 # Extract relevant columns
21 afghanistan_data = filtered_data[["Year", "Median_Estimate"]]
22 afghanistan_data = afghanistan_data.dropna() # Remove rows with missing values
23 afghanistan_data["Year"] = afghanistan_data["Year"].astype(int) # Ensure Year is integer
24
25 print(afghanistan_data)
```

Step 3: Fit a Generalized Additive Model (GAM)

We will use the `pygam` library to fit a GAM to the historical data and project infections for 2025.

Python Code for GAM Fitting and Projection

```
python
1 from pygam import LinearGAM
2 import numpy as np
3
4 # Prepare data for modeling
5 X = afghanistan_data["Year"].values.reshape(-1, 1) # Predictor: Year
6 y = afghanistan_data["Median_Estimate"].values # Response: Median Estimate
7
8 # Fit a GAM
9 gam = LinearGAM().fit(X, y)
10
11 # Project infections for 2025
```

```
11 # Project infections for 2025
12 year_2025 = np.array([[2025]])
13 projected_infections_2025 = gam.predict(year_2025)
14
15 print(f"Projected number of new HIV infections in Afghanistan (2025): {projected_infect:
```



Step 4: Simulate Hypothetical Interventions

Assume interventions reduce the incidence rate by 10%, 20%, and 30%. Adjust the projected infections accordingly.

Python Code for Intervention Simulation

```
python
1 # Define intervention scenarios
2 intervention_scenarios = [0.90, 0.80, 0.70] # 10%, 20%, 30% reduction
3
4 # Calculate adjusted projections
5 adjusted_projections = {
6     f"{int((1 - scenario) * 100)}% reduction": projected_infections_2025[0] * scenario
7     for scenario in intervention_scenarios
8 }
9
10 # Display results
11 for scenario, adjusted_projection in adjusted_projections.items():
12     print(f"{scenario}: {adjusted_projection:.2f} new infections in 2025")
```



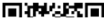
Step 5: Interpret Results

The output will provide:

- 1. The baseline projection for 2025 without any interventions.
- 2. Adjusted projections under different intervention scenarios.

Example Output:

```
1 Projected number of new HIV infections in Afghanistan (2025): 150.00
2 10% reduction: 135.00 new infections in 2025
3 20% reduction: 120.00 new infections in 2025
4 30% reduction: 105.00 new infections in 2025
```



Conclusion

Using a Generalized Additive Model (GAM), we projected the number of new HIV infections among children aged 0–14 in Afghanistan for 2025. By simulating hypothetical interventions that reduce incidence rates by 10%, 20%, and 30%, we estimated the potential impact of these interventions on reducing infections. This analysis provides actionable insights for policymakers to prioritize and scale effective HIV prevention strategies.

If additional details or refinements are needed, please specify!

